

Chad Carmichael

✉ CGC24B@FSU.EDU 📞 954.487.0472 📍 Broward, Florida 🌐 Chad Carmichael 🇺🇸 US Citizen

EDUCATION

Mechanical Engineering

Expected May 2029 | GPA | 3.8

Florida State University

Vires Scholar — merit scholarship, \$16,000 over 8 semesters. | **IB Diploma** recipient, Deerfield Beach High School.

PROFILE

As a freshman mechanical engineering student passionate about robotics and automation, I seek internship or co-op opportunities where I can apply and grow my skills in mechanical systems, CAD, embedded software, and team-based engineering projects. I aim to contribute meaningfully from early on and learn from experienced engineers in real-world settings.

EXPERIENCE/ PROJECTS

TurfTrack / FairLie — Smart Golf Training Mat

- Architected a **3-board embedded sensor system** (ESP32-S3, XIAO nRF52840, Pro Micro nRF52840) fusing **6 pressure sensors**, a 9-axis IMU, and a 60GHz radar module to capture real-time swing and ball-strike data
- **Collaborated directly with contract manufacturers** on PCB fabrication and housing production drawings, translating CAD models into manufacturing-ready documentation
- Engineered a **BLE telemetry pipeline** streaming live sensor data toward an iOS/SwiftUI companion app reporting ball speed, club speed, launch angle, and carry distance

Personal NAS Server

Mini-ITX Build · SolidWorks · Cursor

- **Built a mini-ITX NAS server** with **8TB of storage**, paired with a **custom-written program** (developed in Cursor) that automatically ingests and **organizes photography and coursework files** offloaded from an SSD, eliminating a recurring manual file-sorting bottleneck
- **Designed** the server enclosure in **SolidWorks**

LEADERSHIP AND BUSINESS VENTURES

Co-Founder — EZclickAI & C2x Visuals

- **Scaled client apps and personal brands** to **1M+ combined online views** through short-form content strategy, editing, and distribution
- **Grew one client's revenue by 50% in a single month** by rebuilding their brand presence and content pipeline
- **Collaborate** with FSU Unconquered Media

APP DEVELOPMENT

- **Engineered** and **deployed** a full-stack iOS application using React Native and Expo, **implementing** a swipe-based UI to modernize university discovery.
- **Assisted** in architecting an **automated** data pipeline using the ChatGPT API (LLMs) to clean and normalize a high-volume dataset of **300,000+** raw records, **transforming** them into a structured repository of **1,000+** validated profiles for analysis.

SKILLS

Skills — Programming / Software: C++ , Matlab, Python • **CAD & Design:** Autodesk, SolidWorks (V8 Engine/Stirling Engine Design) • **Productivity Tools:** Microsoft Excel (data analysis, modeling, spreadsheets), • **Core Strengths:** Leadership, Communication, Teamwork, Organization, Time Management, Problem Solving

CLUBS

- National Society of Black Engineers (NSBE) — Member/ NextGen committee
- SMILI (Student Minority in Mentor Leadership Initiative) — Member
- ASCE/FES (American Society of Civil Engineer/ Florida Engineering Society) - 3D Printing Team